TITLE PAGE

Problem Statement ID – Not specified

Problem Statement Title – Digital Farmer Producer Organization (FPO) Management Platform

Theme – Agriculture

PS Category – Software

Team ID – ZZC4L9 Not specified

Team Name – MET26

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IDEA TITLE

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- Digital Farmer Producer Organization (FPO) Management Platform

• Centralized digital platform for farmer registration, procurement, inventory, marketplace, logistics and payments.
• Connects farmers, FPO administrators and buyers through one system.
• AI-assisted price trends and demand forecasting support better selling and stocking decisions.
• Transparent digital records reduce manual work and improve traceability.
• Innovation: end-to-end FPO workflow + AI market intelligence + role-based dashboards.

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TECHNICAL APPROACH

- Technologies

• Frontend: React / HTML-CSS-JS
• Backend: Python (Flask/FastAPI) or Node.js
• Database: MySQL / PostgreSQL
• AI/ML: Python, Pandas, Scikit-learn
• APIs: Market-price / weather data (where available)

Methodology

Farmer → Procurement → Inventory → AI Price/Demand Analysis → Buyer Order → Logistics → Payment → Dashboard

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FEASIBILITY AND VIABILITY

- Feasibility

• Software-first solution; can be developed as a web/mobile prototype.
• Modular architecture allows gradual implementation.
• Uses commonly available open-source development tools.

Challenges & Risks
• Data availability and data quality for AI predictions.
• Farmer digital literacy and language barriers.
• Internet/connectivity limitations.
• Data privacy and transaction security.

Mitigation
• Start with pilot FPO data; validate models continuously.
• Multilingual, simple UI and guided workflows.
• Offline-friendly data capture and secure authentication.

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IMPACT AND BENEFITS

- Target Audience

• Farmer members • FPO administrators • Buyers • Logistics partners

Impact & Benefits

• Faster and more efficient FPO operations
• Better inventory visibility and reduced wastage
• Improved market access and buyer connectivity
• Transparent procurement, orders and payments
• Data-driven market decisions for farmers
• Potential improvement in farmer income and FPO efficiency
• Scalable across crops, FPOs and regions

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RESEARCH AND REFERENCES

- 1. Smart Kopargaon Hackathon 2026 – Official Problem Statement / Presentation Template

2. Department of Agriculture & Farmers Welfare, Government of India – Farmer Producer Organizations
   https://agriwelfare.gov.in/
3. e-NAM – National Agriculture Market
   https://www.enam.gov.in/
4. Small Farmers' Agribusiness Consortium (SFAC)
   https://sfacindia.com/
5. Research areas: FPO management, agricultural market access, inventory and demand forecasting.